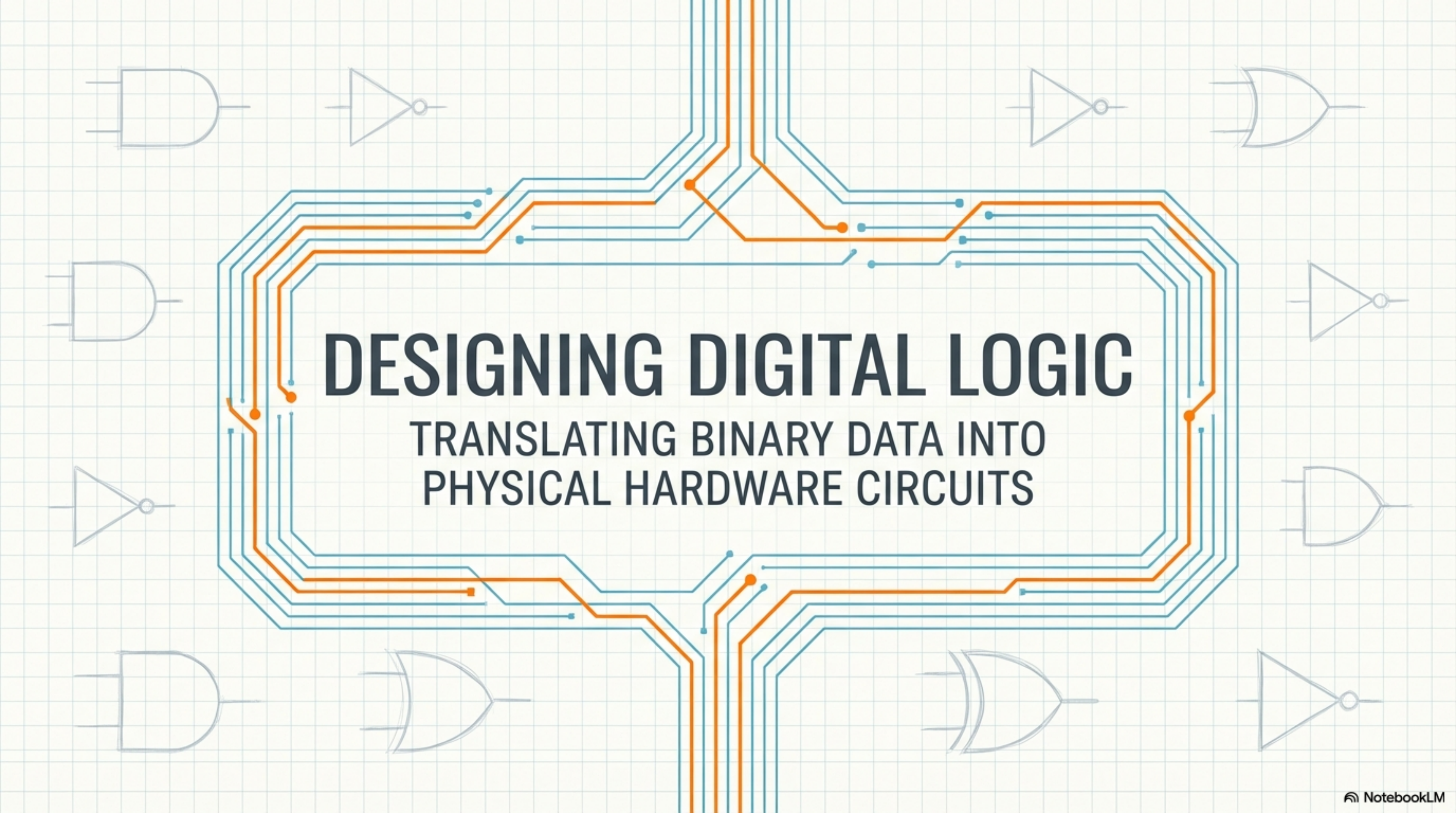


The background of the image is a light gray grid. It features several faint, light blue logic gate symbols: AND gates, OR gates, and NOT gates, scattered around the central text. A complex circuit of blue and orange lines, representing data paths, weaves around the central text area, with some lines ending in small dots.

DESIGNING DIGITAL LOGIC

TRANSLATING BINARY DATA INTO PHYSICAL HARDWARE CIRCUITS

Computers require all human inputs to be converted into binary signals

When a user types the decimal number 25, the system divides it by its binary base (2). Reading the remainders from bottom to top yields the machine-readable sequence: 11001.

$$25 / 2 = 12 \text{ remainder } 1$$

$$12 / 2 = 6 \text{ remainder } 0$$

$$6 / 2 = 3 \text{ remainder } 0$$

$$3 / 2 = 1 \text{ remainder } 1$$

$$1 / 2 = 0 \text{ remainder } 1$$

Read Bottom to Top

Binary numbers are simply a mathematical representation of physical voltage
Electronic devices process information by reading the presence or absence of electrical current.



0 Volts
= Low
= OFF
= False
= 0

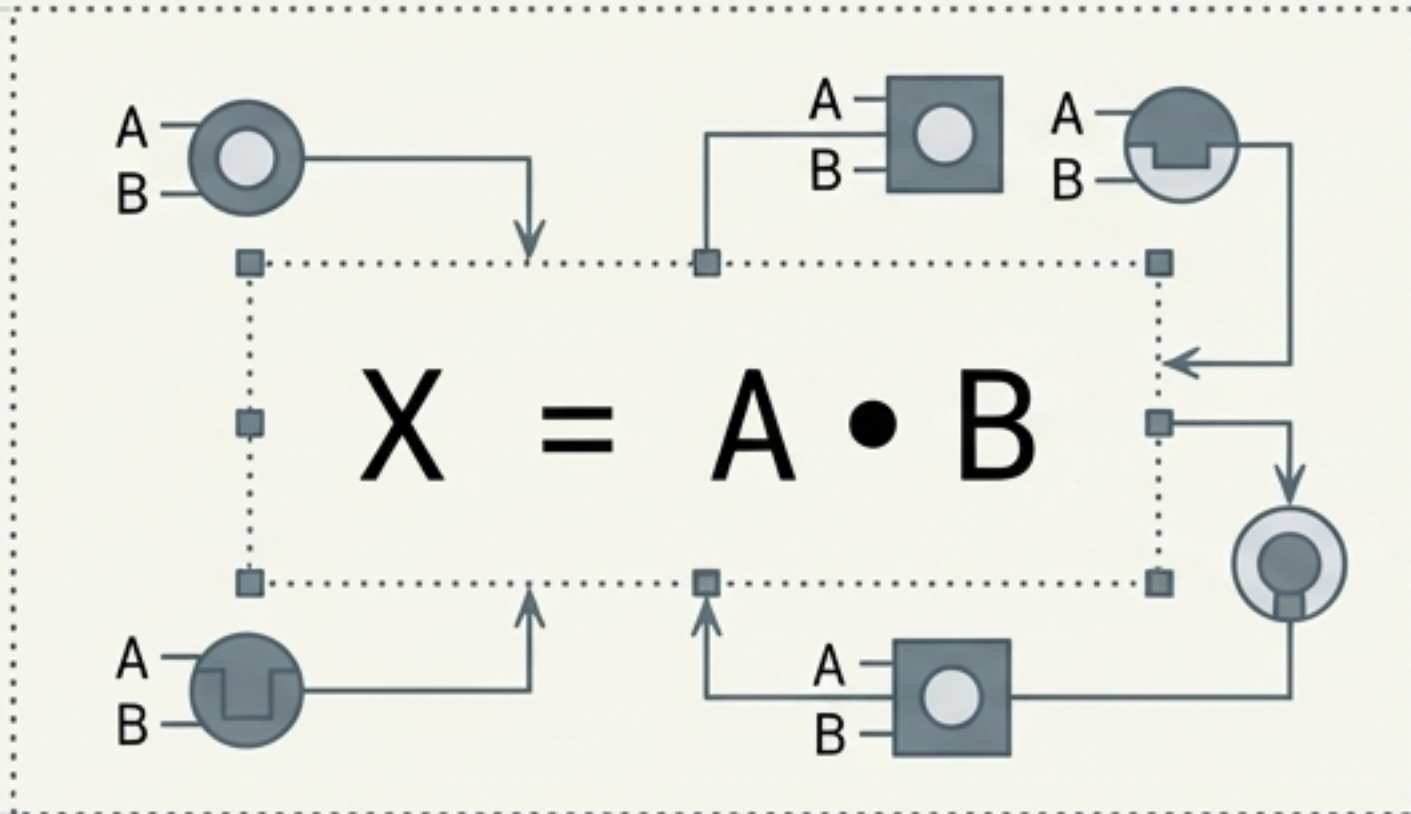


5 Volts
= High
= ON
= True
= 1

Boolean logic is the math; Logic gates are the physical machinery

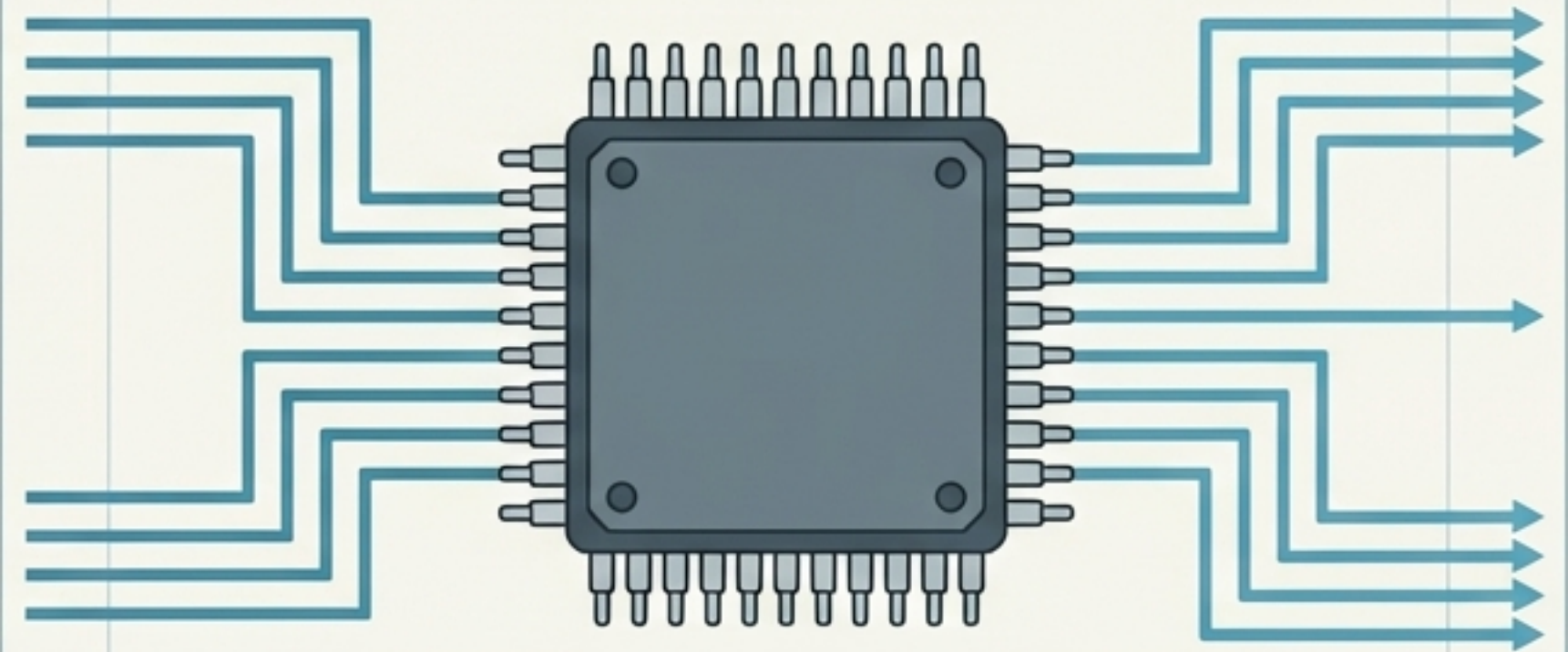
Boolean Operations

Boolean Algebra is the branch of mathematics that evaluates logical states (True/False).



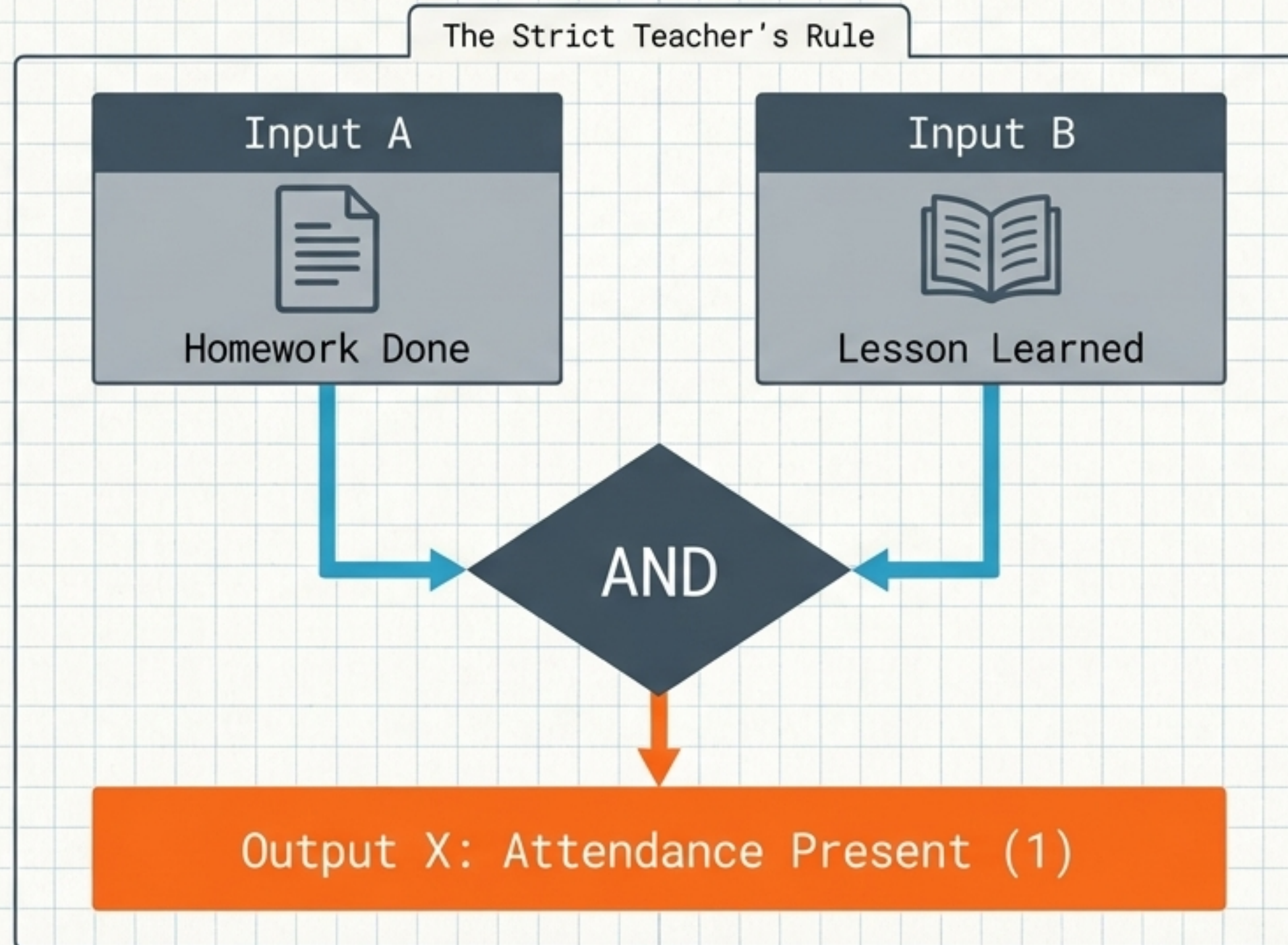
Logic Gates

Logic Gates are the physical electronic devices inside circuits that actively enforce those rules to direct electrical currents.



The AND Operation requires all conditions to be met

Imagine a strict teacher evaluating a student. To be marked Present (1), the student must have their Homework done (Input A) AND their Lesson learned (Input B). If either is missing (0), the student is marked Absent (0).



Engineering the AND Logic Gate

In an AND circuit, switches are placed in series. If even one switch is open (0), the pathway is broken, and the output is zero.

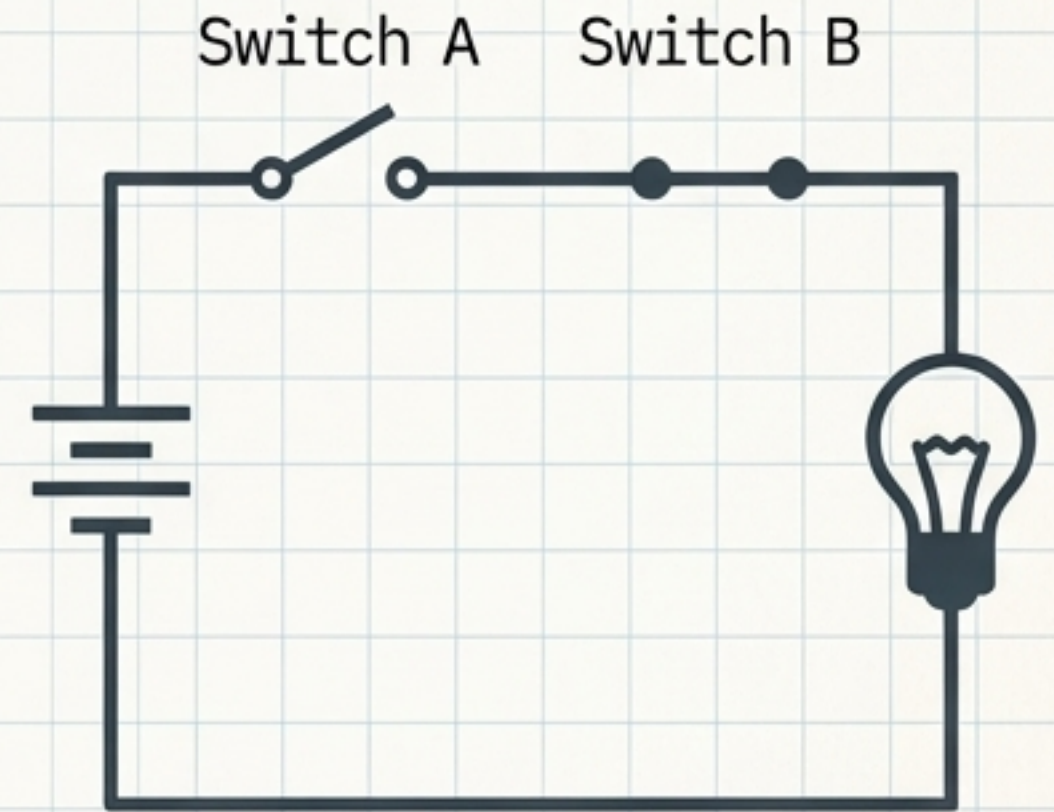
The Math

$$X = A \cdot B$$

The Truth Table

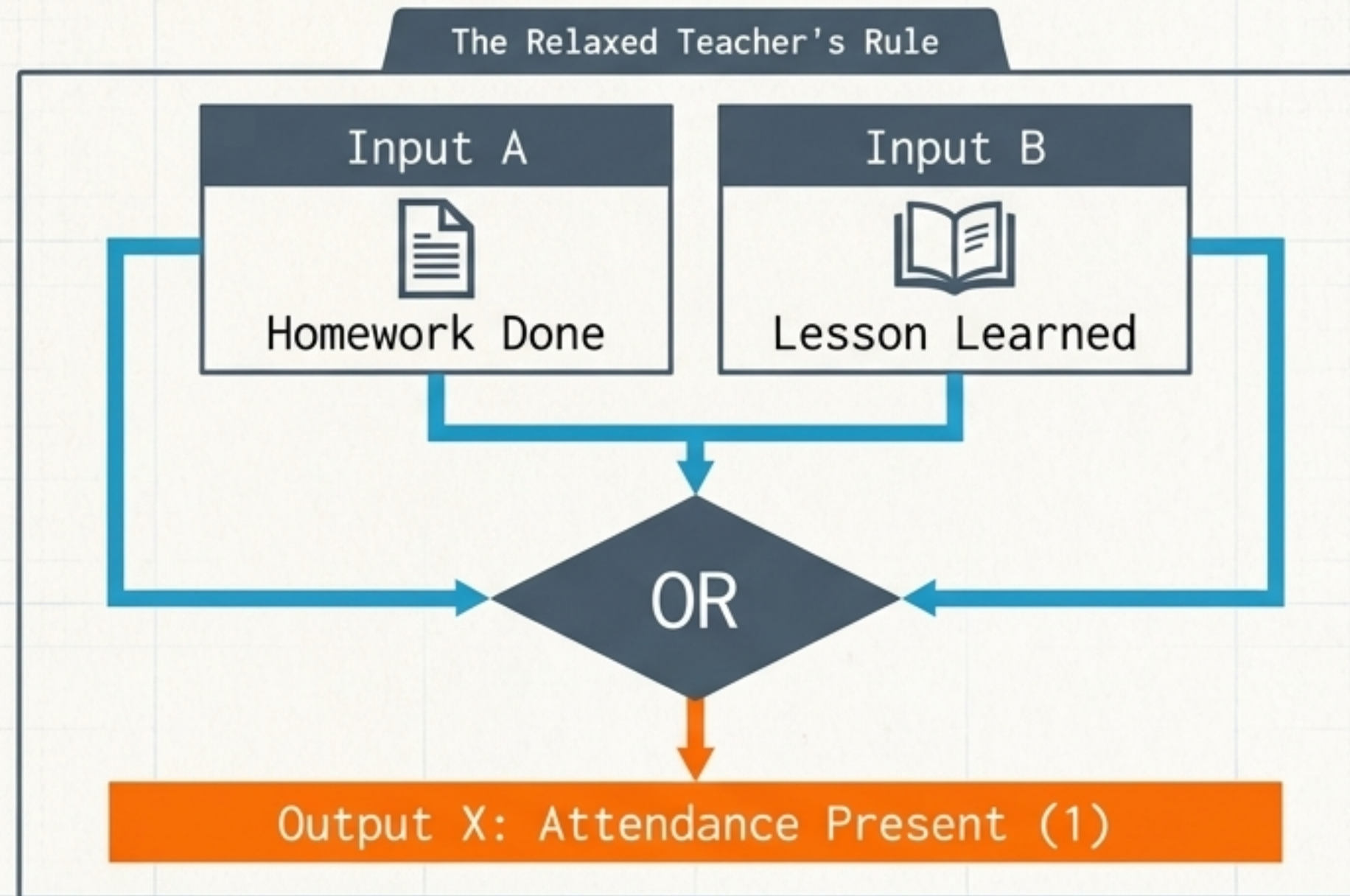
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

The Circuit



The OR Operation requires at least one condition to be met

Now the teacher relaxes the rules. A student will be marked Present (1) if they have EITHER their Homework done (A) OR their Lesson learned (B). They are only marked Absent if they did neither.



Engineering the OR Logic Gate

In an OR circuit, switches are placed on parallel tracks. Current bypasses any open switch (0) as long as an alternate closed switch (1) is available.

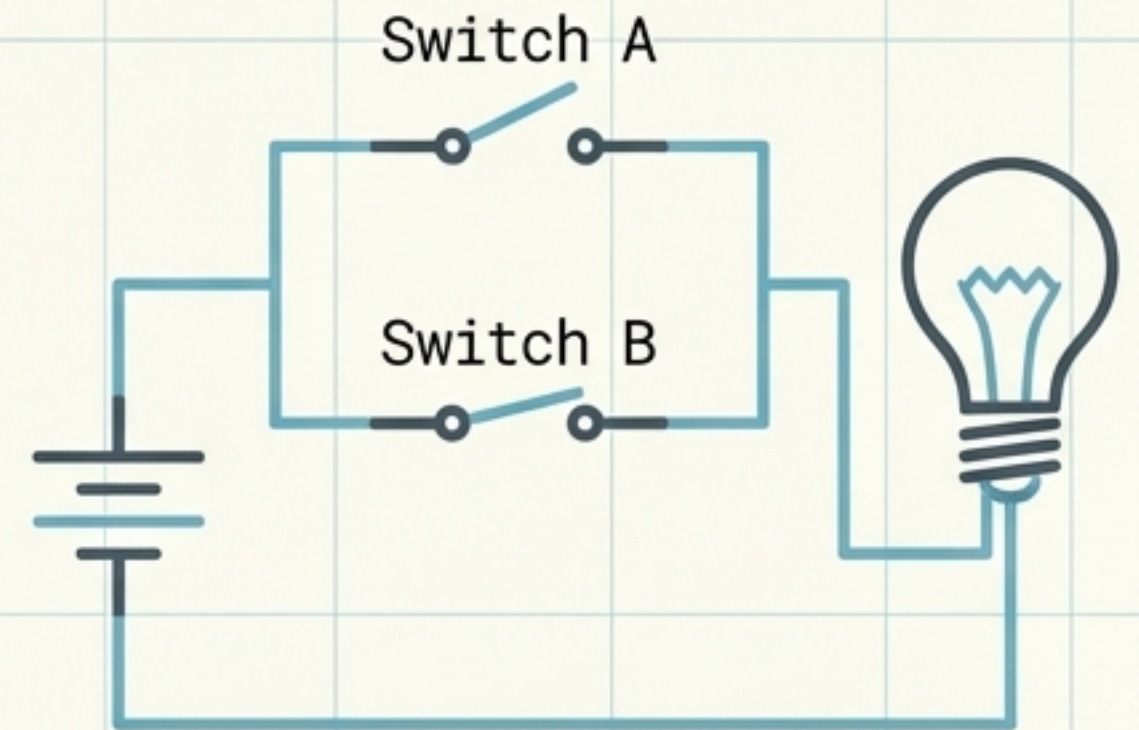
The Math

$$X = A + B$$

The Truth Table

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

The Circuit



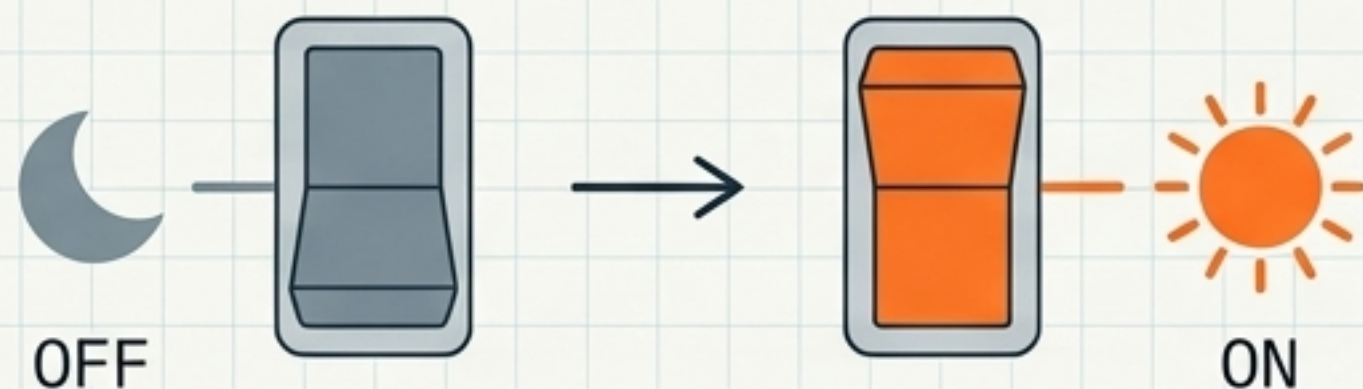
The NOT Gate actively reverses the input state

Unlike AND/OR, the NOT gate only takes a single input. It acts as a strict inverter. If a high voltage (1) enters, a low voltage (0) exits.

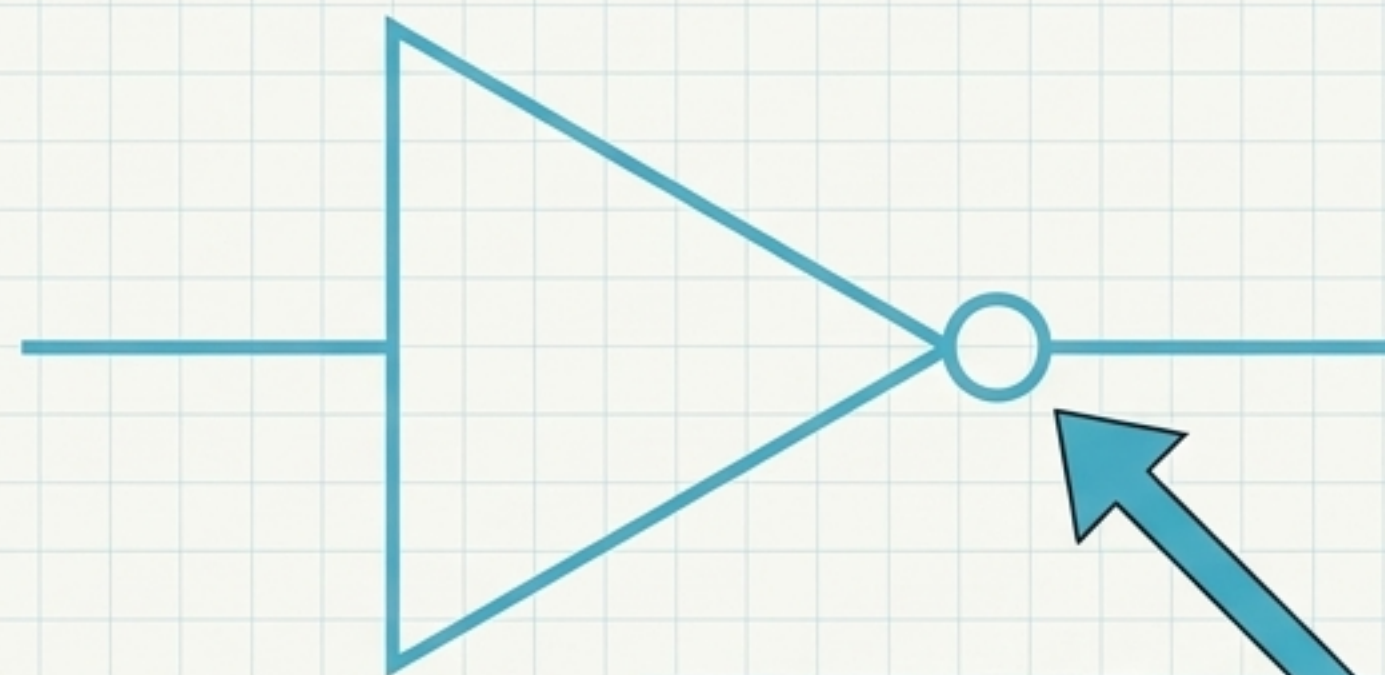
Math & Concept

$$X = A'$$

Input (A)		Output (X)
0	->	1
1	->	0



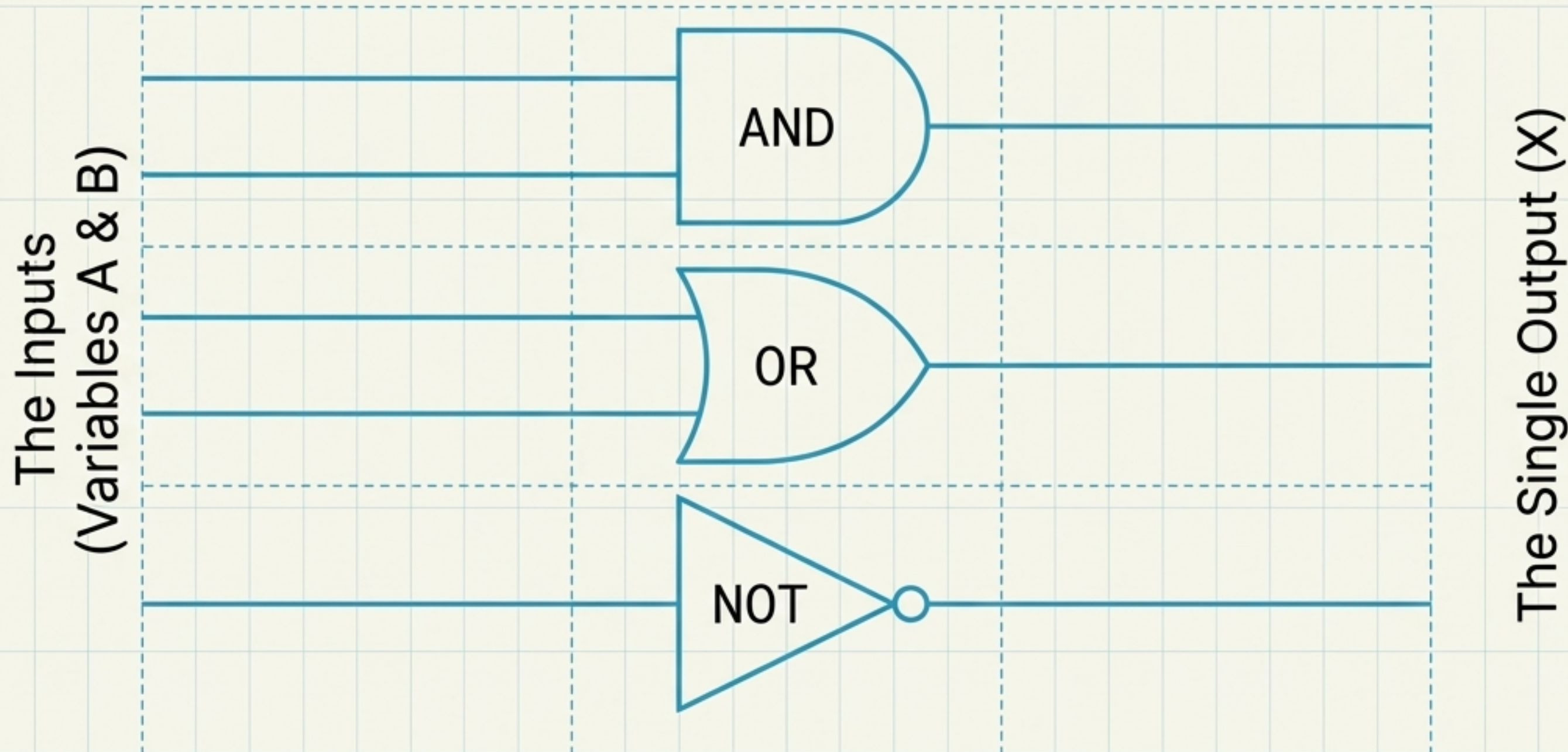
The Anatomy



The Inversion Bubble: This circle physically represents the reversal of the signal.

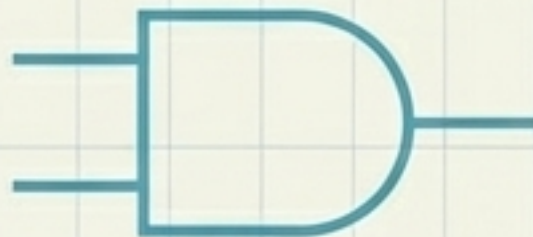
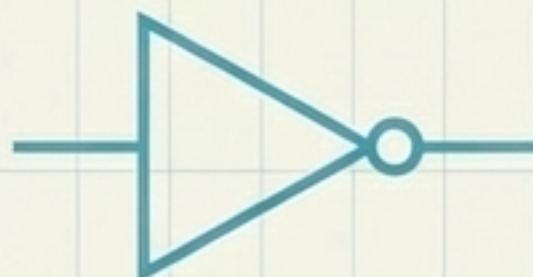
Visual Anatomy of Logic Gates

Physical gate devices are standardized by unique geometric shapes in technical schematics. The left side always receives the input signals, the center shape dictates the logical operation, and the right side pushes the final electrical output.



Chapter 3 Synthesis: Logic Gate Master Reference

Combine these foundational hardware gates to build complex digital computer networks. Use this matrix to verify truth tables and boolean expressions during circuit analysis.

Gate Name	Boolean Expression	The Rule	Logic Symbol	Diagnostic Truth Table																				
AND	$X = A \cdot B$	Both must be 1		<table><tr><td>A</td><td>B</td><td>-></td><td>X</td></tr><tr><td>0</td><td>0</td><td>-></td><td>0</td></tr><tr><td>0</td><td>1</td><td>-></td><td>0</td></tr><tr><td>1</td><td>0</td><td>-></td><td>0</td></tr><tr><td>1</td><td>1</td><td>-></td><td>1</td></tr></table>	A	B	->	X	0	0	->	0	0	1	->	0	1	0	->	0	1	1	->	1
A	B	->	X																					
0	0	->	0																					
0	1	->	0																					
1	0	->	0																					
1	1	->	1																					
OR	$X = A + B$	At least one must be 1		<table><tr><td>A</td><td>B</td><td>-></td><td>X</td></tr><tr><td>0</td><td>0</td><td>-></td><td>0</td></tr><tr><td>0</td><td>1</td><td>-></td><td>1</td></tr><tr><td>1</td><td>0</td><td>-></td><td>1</td></tr><tr><td>1</td><td>1</td><td>-></td><td>1</td></tr></table>	A	B	->	X	0	0	->	0	0	1	->	1	1	0	->	1	1	1	->	1
A	B	->	X																					
0	0	->	0																					
0	1	->	1																					
1	0	->	1																					
1	1	->	1																					
NOT	$X = A'$	Reverses the input		<table><tr><td>A</td><td>-></td><td>X</td></tr><tr><td>0</td><td>-></td><td>1</td></tr><tr><td>1</td><td>-></td><td>0</td></tr></table>	A	->	X	0	->	1	1	->	0											
A	->	X																						
0	->	1																						
1	->	0																						